

Diffusion-Based Generation of Novel Pokémon

with Conditional Evolution Modeling

MSML 612 - Final Presentation

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Problem Motivation

Goal: Generate novel, visually coherent Pokémon designs under structured conditional controls.

Challenge	Why It Matters
Multi-source style alignment	Official art, game sprites, and 3D renders must correspond
Attribute conditioning	Type, evolution stage, and style must guide generation
Evolution chain coherence	Generated forms must remain visually related across a chain
Domain-specific structure	Pokémon generation is not just image synthesis: it is structured conditional generation

Standard unconditional image generators cannot capture evolution progression or type-based stylistic cues.

From Baseline to Conditional Diffusion

Why move past the GAN baseline?

- The preliminary image-to-image GAN captured broad **color palettes** and rough **silhouettes** of target sprites
- It struggled with **fine-grained detail**, **sharp edges**, and **mode coverage** (outputs collapsed toward average sprites)
- It was fundamentally a **translator**, not a **generator**: it could only replay styles it had been shown paired with

What a conditional diffusion model gives us instead

- **Iterative denoising** produces sharper, higher-frequency detail than a single-pass GAN generator
- Naturally supports **structured conditioning** (type, style, stage, reference image) via FiLM + cross-attention
- Supports **true generation** of unseen Pokémon, not just translation of existing ones
- Training is more **stable** (no adversarial collapse) on our relatively small (~2k–6k) dataset

Dataset Overview

Three complementary image sources, now unified into a single **diffusion-ready** training set with type, stage, and style labels.

Source	Files	Role
SugimoriSprites	1,848	High-quality 2D reference art
PokeSprite	2,853	Game-style sprite (primary target domain)
ProjectPokemon 3D	2,799	Alternate visual domain for cross-format variety

Diffusion-ready mappings (with `types` + `evolution_stage` + `art_style`):

Mapping File	Entries	Role
<code>safe_sprite_to_sprite_types_evolution.json</code>	1,995	Sprite → Sprite (+ prev-evo image)
<code>safe_sugimori_to_sugimori_types_evolution.json</code>	1,674	Sugimori → Sugimori (+ prev-evo image)
<code>safe_project_to_project_types_evolution.json</code>	2,704	3D → 3D (+ prev-evo image)
Total training entries	6,373	Merged in <code>PokemonDiffusionDataset</code>

Data Sources: Visual Comparison

Style	Bulbasaur	Charizard
SugimoriSprites (Official Art)		
PokeSprite (Game Sprite)		
ProjectPokemon 3D		

Each source captures a different visual style of the **same** Pokémon: cross-format alignment is critical.

Dataset Enrichment: Types + Evolution Stage

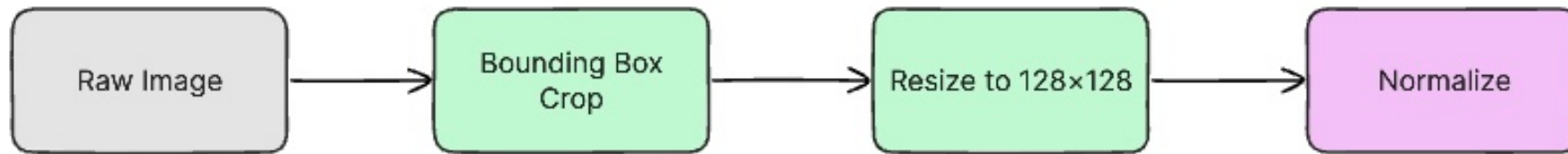
Per-entry schema (JSON):

```
{
  "prev": null,
  "next": "bulbasaur",
  "prev_sprite": null,
  "next_sprite": "poke-data/PokeSprite/0001bulbasaur/bulbasaur.png",
  "types": ["grass", "poison"],
  "evolution_stage": "base",
  "art_style": "sprite"
}
{
  "prev": "bulbasaur",
  "next": "ivysaur",
  "prev_sprite": "poke-data/PokeSprite/0001bulbasaur/bulbasaur.png",
  "next_sprite": "poke-data/PokeSprite/0002ivysaur/ivysaur.png",
  "types": ["grass", "poison"],
  "evolution_stage": "evo 1",
  "art_style": "sprite"
}
```

Curation challenges we had to solve (not off-the-shelf):

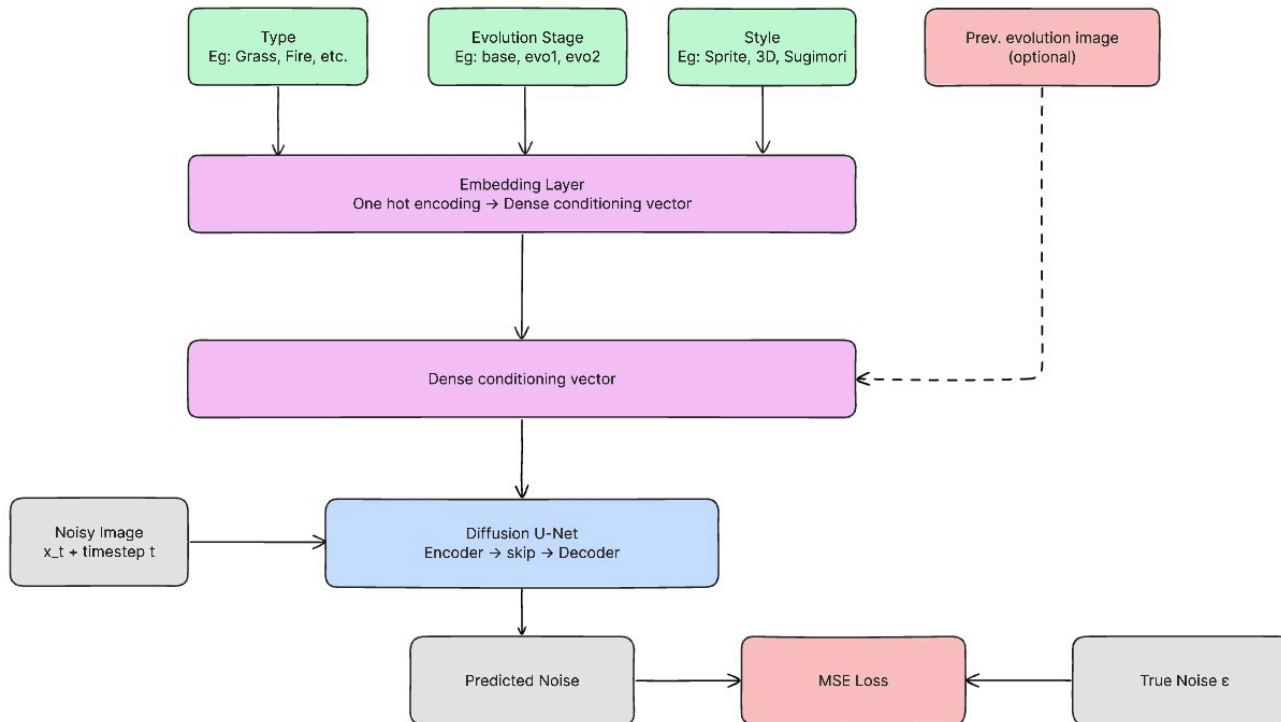
- **Cross-format name collisions:** same Pokémon, different filename conventions across sources (e.g. `0006 Charizard.png` vs `charizard.png` vs `poke_capture_0006_*.png`); resolved via normalized lookup against `poke-data/pokedex.json`
- **Regional/cosmetic variants:** `meowth-galar`, `bulbasaur shiny`, mega, gigantamax share base IDs; stripped and tagged so type lookup stays unambiguous
- **Evolution-stage resolution:** no source labels stage directly; derived via **BFS over `prev` → `next` edges** from chain roots, with **PokeAPI fallback**
- **Art-style labelling:** tagged per mapping file (`sprite`, `sugimori`, `3d`) so one unified dataset drives **style-conditioned** sampling

Preprocessing Pipeline



Filename normalization → form-variant handling (Mega, Gigantamax) → cross-format mapping generation → **types + evolution enrichment** → RGBA 64×64 resize for diffusion training

System Architecture: Conditional U-Net Diffusion



Design decisions under constraint:

- **Small dataset (~6K) → diffusion over GAN:** DDPM training is stable in low-data regimes where adversarial training collapses
- **Multi-modal conditioning** (18 types $\times$ 3 stages $\times$ 3 styles + a *reference image*) → simple concatenation is insufficient; we use **FiLM (γ, β) modulation at every ResBlock** plus a dedicated CNN encoder for the prev-evo image
- **Self-attention at 16 $\times$ 16 and 8 $\times$ 8:** captures long-range shape coherence that pure convolutions miss at this resolution
- **RGBA-preserving pipeline:** 4-channel inputs throughout (non-trivial: standard ImageNet stats break alpha); keeps sprite transparency intact end-to-end
- **Unified conditioning vector:** $[t_emb \parallel c_emb \parallel p_emb]$ → MLP → cond, shared across all U-Net levels

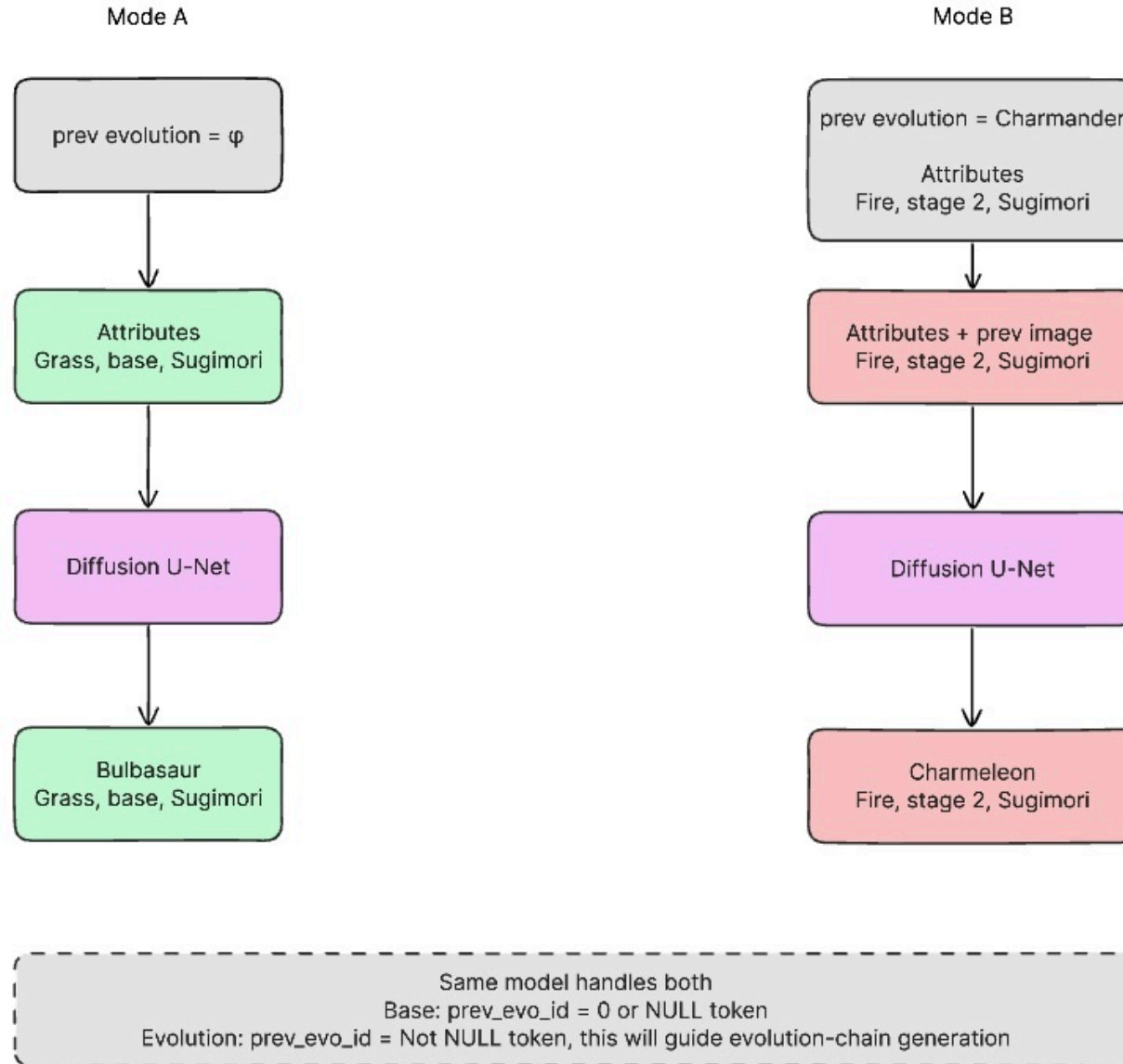
Architecture: Conditioning Design

Five heterogeneous conditioning signals: **categorical**, **set-valued**, **continuous**, and **image-valued**, all fused into a single control vector.

Conditioning Input	Encoding	Why this encoding
Pokémon Type (e.g., Fire, Water)	Multi-hot over 18 types (supports dual-type)	One-hot would lose dual-typing entirely
Evolution Stage (base / evo1 / evo2)	One-hot over 3 stages	Discrete structural complexity control
Visual Style (3d / sugimori / sprite)	One-hot over 3 styles	Enables cross-domain sampling at inference
Previous Evolution Image	CNN encoder → 128-d vector (masked if absent)	Base-stage Pokémon have no prior; mask token keeps batch shape valid
Timestep <code>t</code>	Sinusoidal embedding → MLP	Standard DDPM timestep encoding

All embeddings are concatenated and fused by `cond_combine` into a single 128-dim vector, which drives **FiLM** (γ , β) **modulation** inside every ResBlock at every resolution, not just at the bottleneck.

Training Procedure



Training Details

Component	Setting
Model	Conditional U-Net, FiLM-modulated, self-attention at 16 & 8
Parameters	~XX M
Dataset	6,373 entries merged across 3d / sugimori / sprite mappings
Hardware	G4 High-RAM GPU: NVIDIA RTX PRO 6000 Blackwell (Google Colab)
Training time	XX hours for XX epochs
Final training loss	XX

TODO: fill in parameter count, epochs completed, wall-clock training time, and final loss once the run is finished.

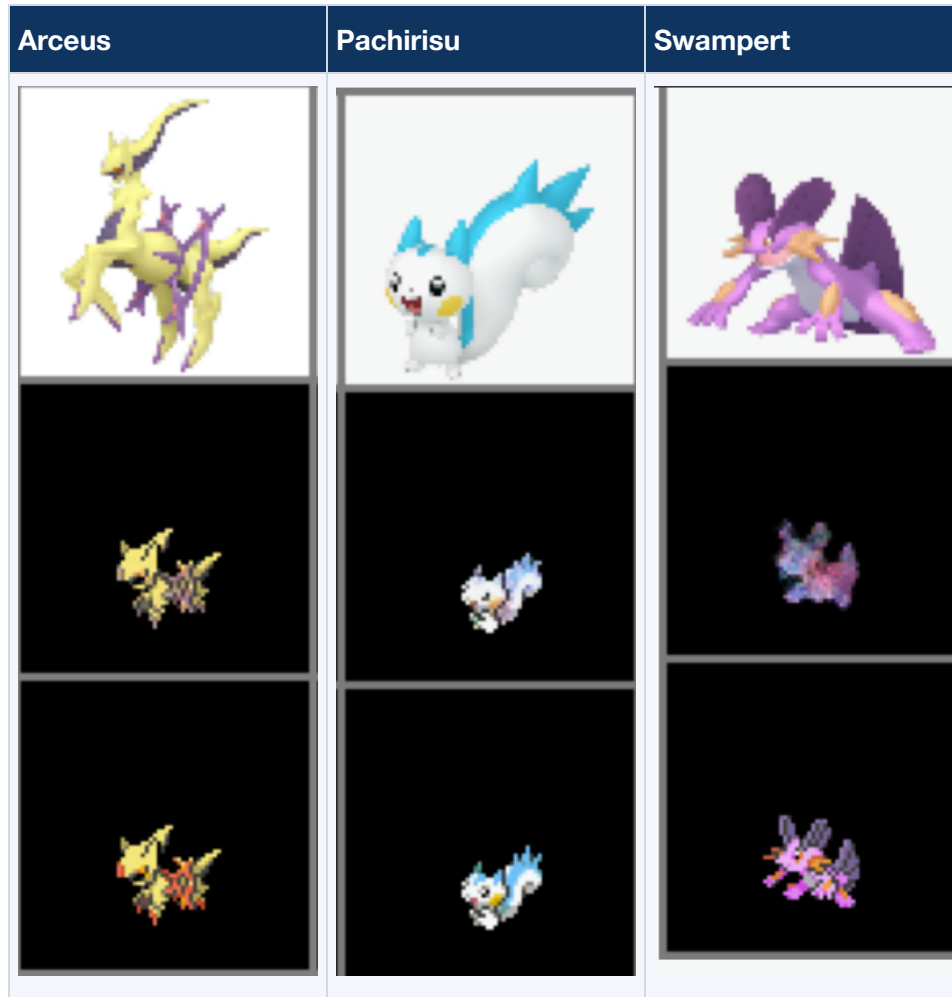
Reproducibility & Code Organization

The full pipeline reproduces end-to-end from a clean checkout with a **single command**.

Concern	How we handle it
Deterministic runs	Fixed seeds for <code>torch</code> , <code>numpy</code> , <code>random</code> ; deterministic <code>DataLoader</code> workers
Dataset splits	Train / val indices saved to JSON; same split across every experiment
Configuration	YAML-driven (<code>configs/final.yaml</code>); all hyperparameters live outside code
Data loading	Single <code>PokemonDiffusionDataset</code> class; unifies all 3 mapping files
Entry point	<code>python train.py --config configs/final.yaml</code> reproduces the full run
Environment	Pinned <code>requirements.txt</code> ; tested on NVIDIA G4 (Colab) and local CUDA 12
Checkpoints	EMA + raw weights saved every N epochs; resumable via <code>--resume</code>
Sampling	<code>python sample.py --ckpt ... --type fire --stage base --style sprite</code>

No hidden manual steps: the mappings, splits, training run, and sampling grids used in this deck are all generated by scripts checked into the repo.

Preliminary GAN Baseline (for comparison)



Top: Input artwork

Middle: GAN-predicted sprite

Bottom: Ground-truth sprite

GAN captures color palettes and rough silhouettes but lacks fine detail and sharpness.

The conditional diffusion model should improve global consistency and recover sharper fine-grained detail.

Diffusion Results: Unconditional Samples

TODO: add a grid of unconditionally-sampled Pokémon from the trained diffusion model. Suggested layout: 4×4 grid of 64×64 RGBA samples on a checkerboard background. Filename suggestion:

`images/diff_uncond_grid.png` .

Diffusion Results: Type-Conditioned Samples

*TODO: add a figure showing samples conditioned on different **types** (e.g. one row per type: fire, water, grass, electric, psychic, dragon), all with `style = sprite`, `stage = base`. Filename suggestion: `images/diff_type_grid.png`.*

Diffusion Results: Style-Conditioned Samples

*TODO: add a figure showing the **same conditioning vector** rendered under each of the three styles (`sprite` , `sugimori` , `3d`) to demonstrate that style conditioning actually transfers across domains. Filename suggestion: `images/diff_style_grid.png` .*

Diffusion Results: Evolution-Chain Generation

*TODO: add horizontal strips from `generate_evolution_chain(...)` showing **base** → **evo1** → **evo2**, where each stage is conditioned on the previous generated image. Show 3–4 chains (e.g. fire, water, grass, dragon). Filename suggestion: `images/diff_evo_chains.png`.*

Diffusion vs. GAN: Side-by-Side

TODO: pick 3 Pokémon and show a 3-row comparison:

Row 1: input artwork / target

Row 2: GAN-predicted sprite (from baseline)

Row 3: Diffusion-generated sprite (same conditioning)

Filename suggestion: `images/diff_vs_gan.png` .

Evaluation Plan

Metric	Description	What It Measures
FID (Fréchet Inception Distance)	Distribution distance between generated and real images	Realism & diversity
SSIM	Structural similarity for paired comparisons	Pixel-level fidelity
Diversity Score	Pairwise distance across generated samples	Mode coverage
Qualitative Review	Side-by-side visual inspection	Attribute adherence

Quantitative Results

Metric	GAN Baseline	Diffusion (ours)
FID ↓	XX	XX
SSIM ↑	XX	XX
Diversity ↑	XX	XX

TODO: fill in the numerical results once evaluation scripts are run on the held-out Pokémon split. Also include per-style FID (sprite / sugimori / 3d) if time allows.

Runtime & Efficiency

Performance is not just sample quality; the rubric also credits **running time** and practical cost.

Measurement	GAN Baseline	Diffusion (ours)
Parameters	XX M	XX M
Peak GPU memory (training, batch=32)	XX GB	XX GB
Wall-clock training time	XX h	XX h
Inference time per sample (T4, 1000 steps)	n/a	XX s
Inference throughput (samples / min)	XX	XX

TODO: log these numbers directly from the training/sampling scripts so they are reproducible from the checkpoint.

Discussion

What worked well

- Unified multi-source dataset (~6.4K entries) with clean `types` / `stage` / `style` labels
- FiLM conditioning integrates cleanly with the U-Net; no architectural surprises
- RGBA-preserving pipeline keeps sprite transparency intact

What was hard

- Cross-format name normalization (shiny, mega, gigantamax, regional variants)
- Evolution-stage resolution required BFS + PokeAPI fallback
- Small dataset relative to typical diffusion training → careful augmentation & EMA were important

Limitations & Future Work

Current limitations

- Trained at **64×64 RGBA**; fine sprite detail still bounded by resolution
- Only 3 styles and 3 evolution stages; real Pokémon have far more visual variants
- No text conditioning (e.g. natural-language prompts for abilities or lore)

Future directions

- **Scale up resolution** with a latent diffusion / super-resolution stage
- **Classifier-free guidance** via conditioning dropout for stronger attribute adherence
- **Text conditioning** using a pretrained text encoder (e.g. CLIP) for descriptive generation
- **Full evolution-chain consistency loss** that ties together all generated stages jointly

Conclusion

- Moved from a **preliminary GAN translator** to a **conditional diffusion generator**
- Built a **6.4K-entry** diffusion-ready dataset with type, stage, and style metadata across sprite, Sugimori, and 3D sources
- Implemented a **FiLM-conditioned U-Net** with a dedicated prev-evolution image encoder
- Demonstrated type-, style-, stage-, and evolution-chain-conditioned generation of novel Pokémon

Conditional diffusion is a substantially better fit than the GAN baseline for the structured, low-data, multi-domain setting of Pokémon generation.

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